

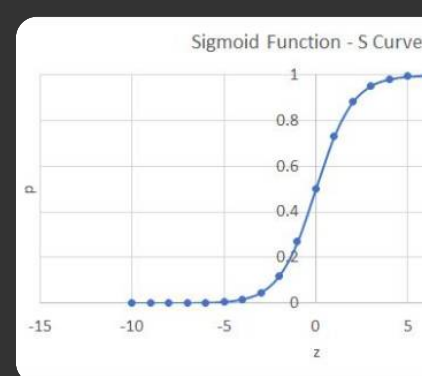
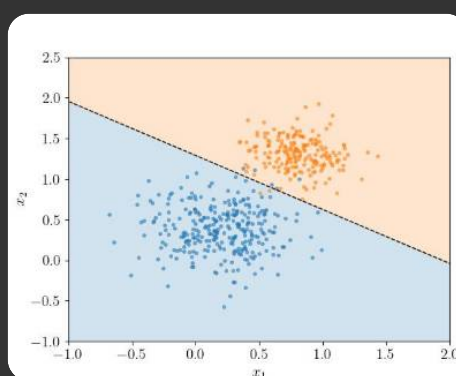
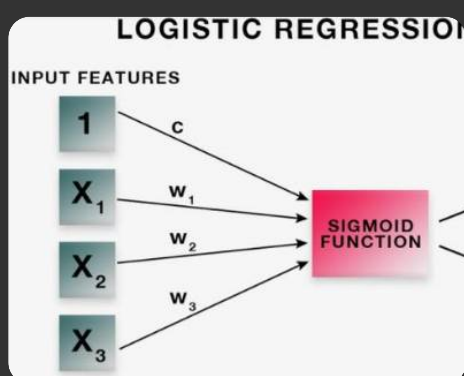
Top ML Models for classification & regression

top classification and regression models

Here's a clean, exam-ready overview of the **top classification and regression models** used in machine learning 🙌

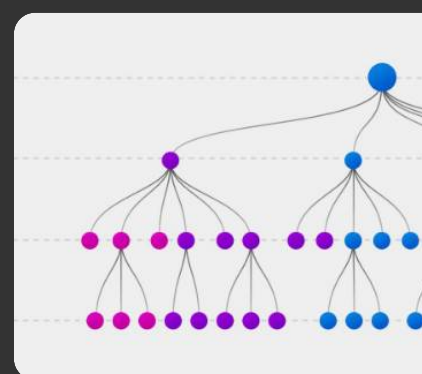
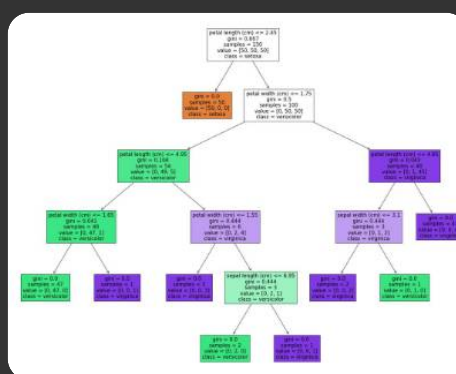
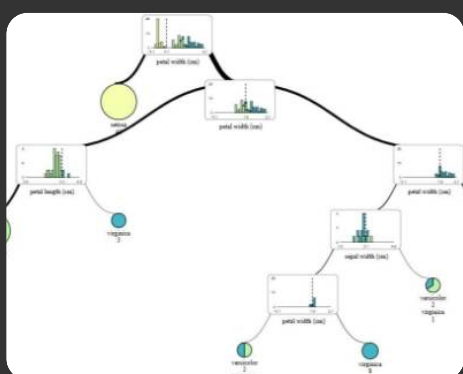
◆ Classification Models (Predict Categories / Labels)

1. Logistic Regression



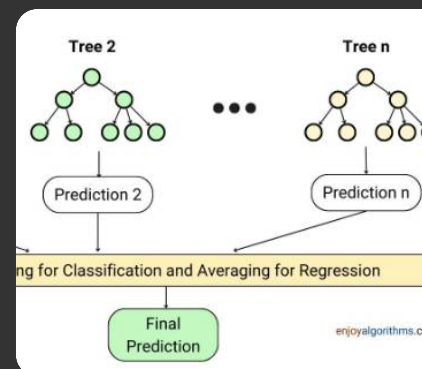
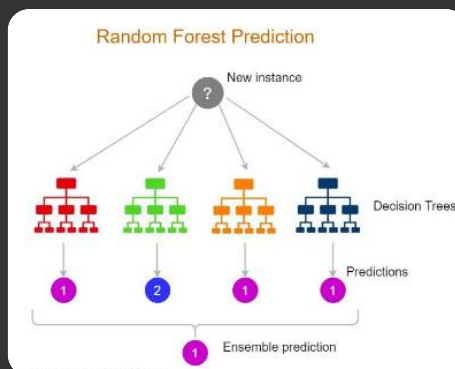
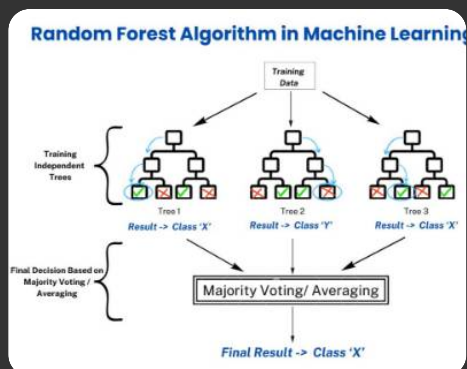
- Used for **binary classification** (Yes/No, 0/1)
- Uses **sigmoid function** to map values between 0 and 1
- Simple, fast, and interpretable

2. Decision Tree



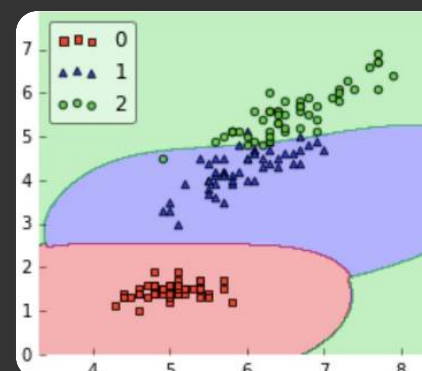
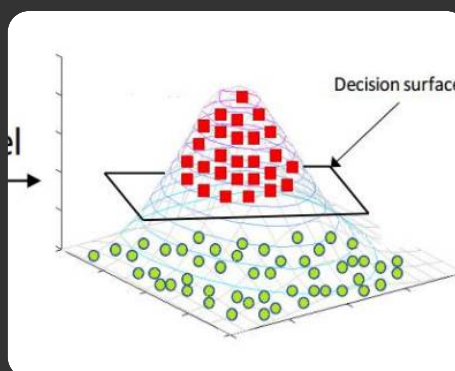
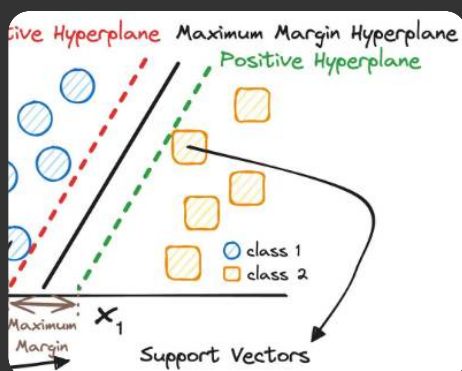
- Splits data into branches based on conditions
- Easy to understand and visualize
- Can handle both numerical & categorical data

3. Random Forest



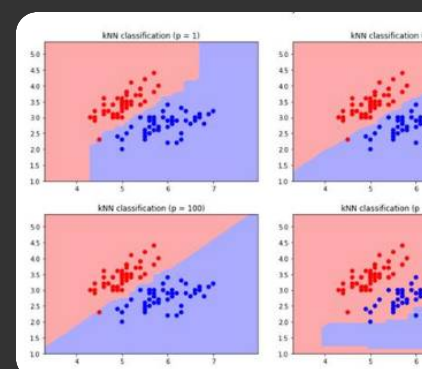
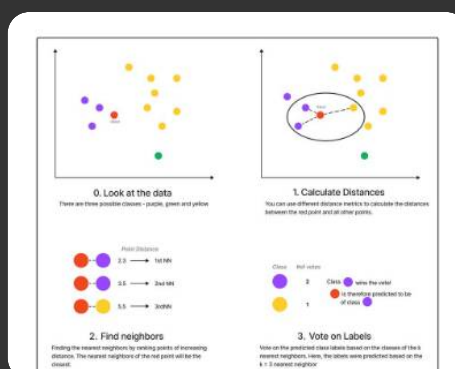
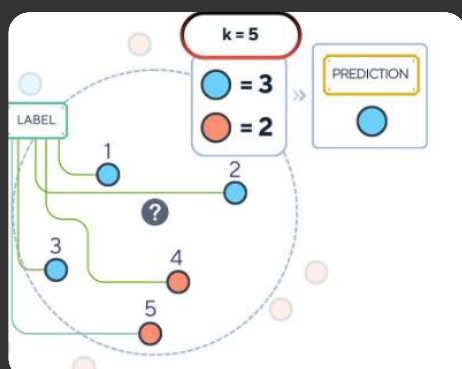
- Ensemble of multiple decision trees
- Uses **majority voting**
- Reduces overfitting → more accurate than a single tree

4. Support Vector Machine (SVM)



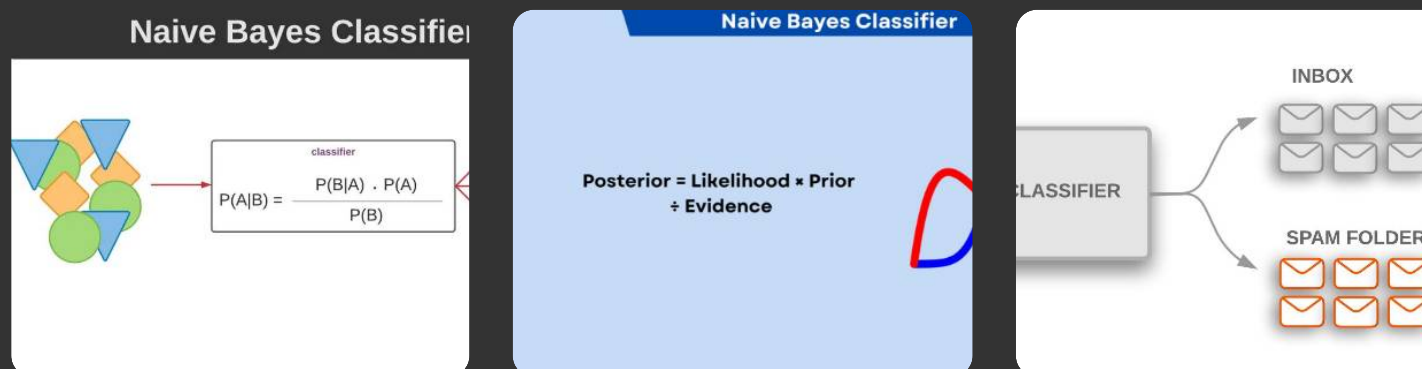
- Finds **optimal hyperplane** to separate classes
- Works well for high-dimensional data
- Can handle non-linear data using kernels

5. K-Nearest Neighbors (KNN)



- Classifies based on nearest data points
- Simple and intuitive
- Slow for large datasets

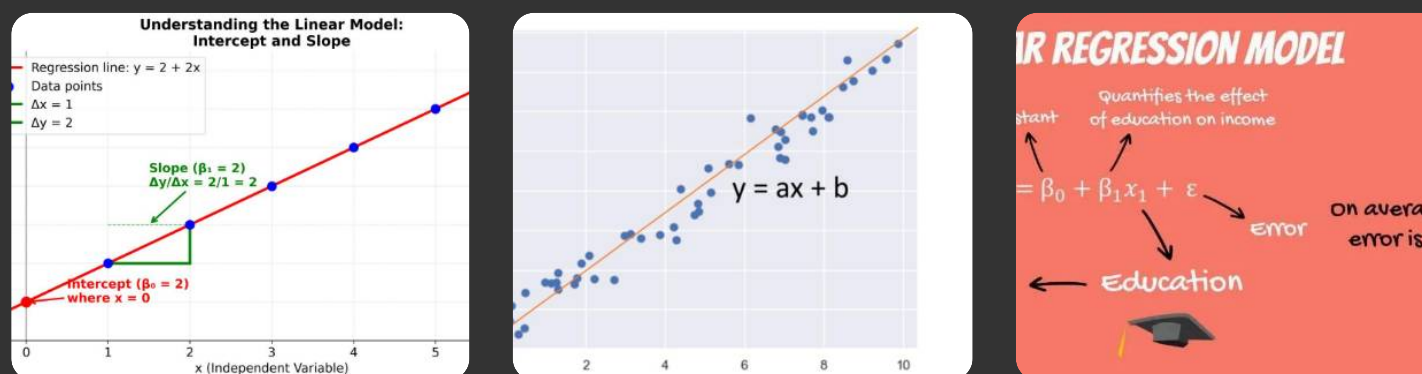
6. Naive Bayes



- Based on **Bayes' Theorem**
- Assumes features are independent
- Very fast, used in spam filtering

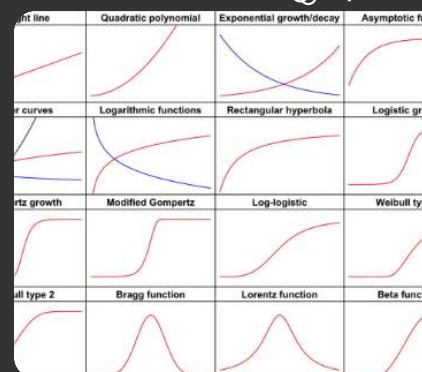
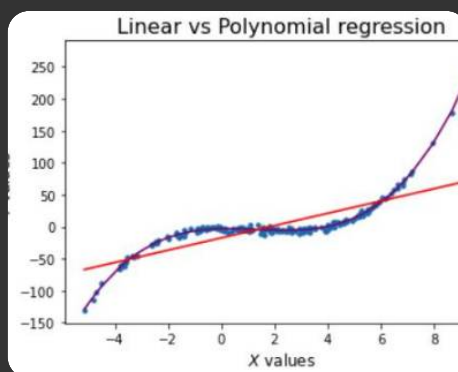
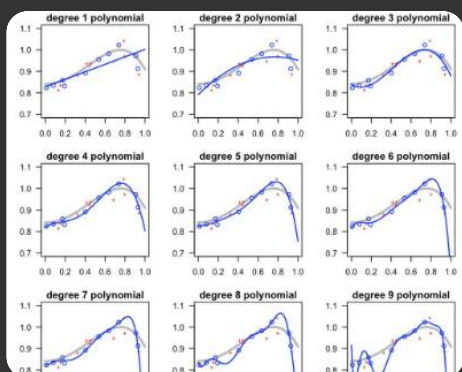
◆ Regression Models (Predict Continuous Values)

1. Linear Regression



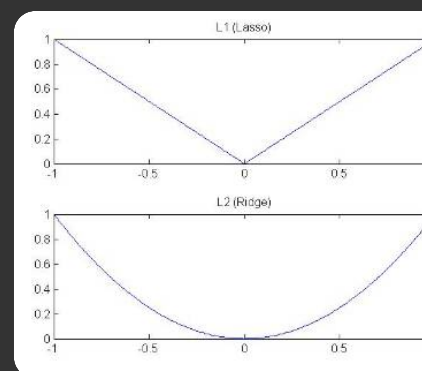
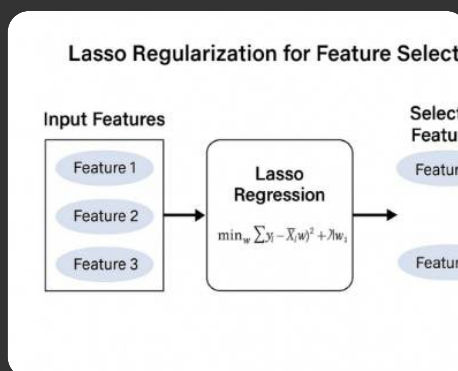
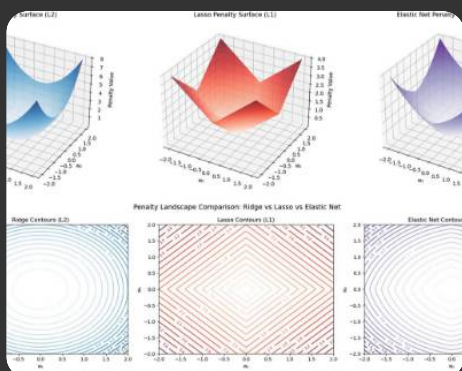
- Predicts continuous values (e.g., price)
- Fits a **straight line**
- Easy and widely used

2. Polynomial Regression



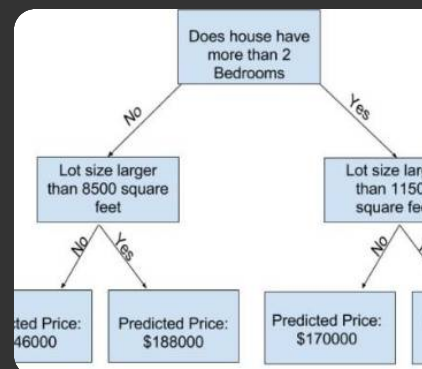
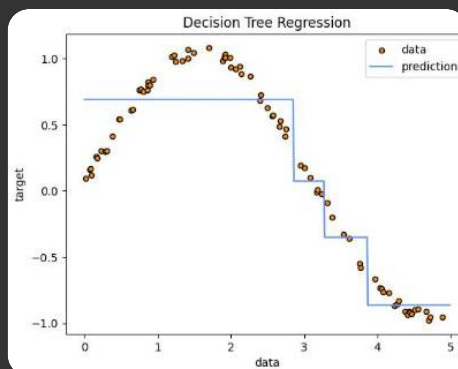
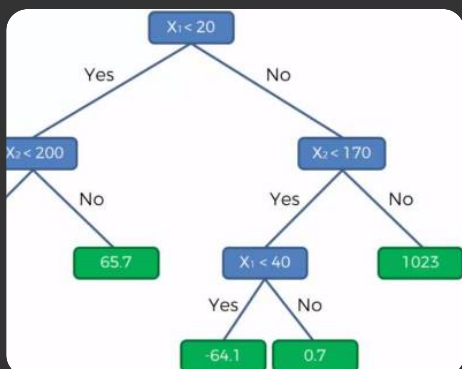
- Extension of linear regression
- Fits a **curved relationship**
- Useful for non-linear data

3. Ridge & Lasso Regression



- Add **regularization** to avoid overfitting
- Ridge → shrinks coefficients
- Lasso → can set some coefficients to zero

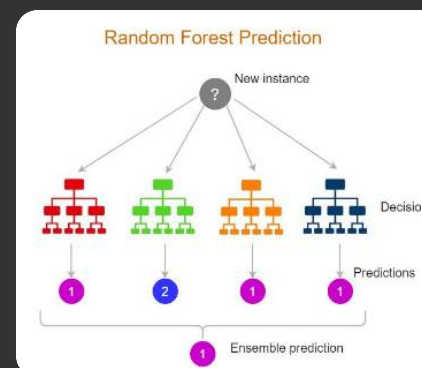
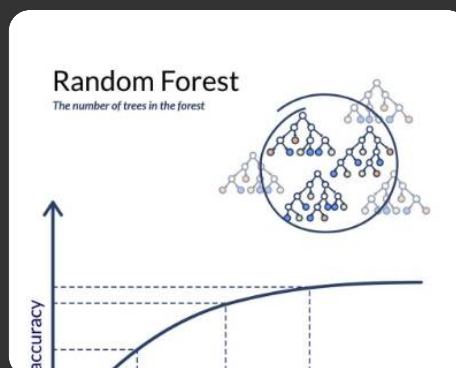
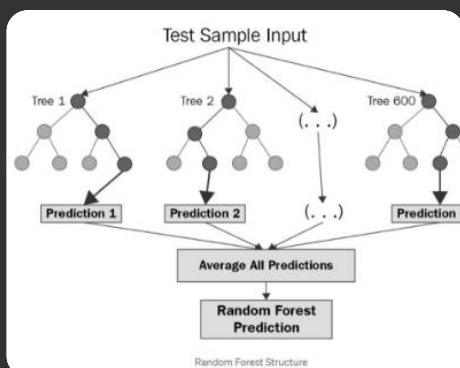
4. Decision Tree Regression



- Splits data into regions
- Predicts average value in each region

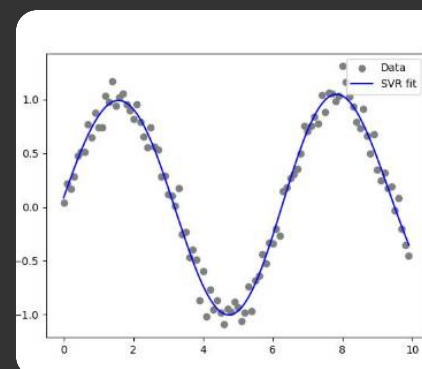
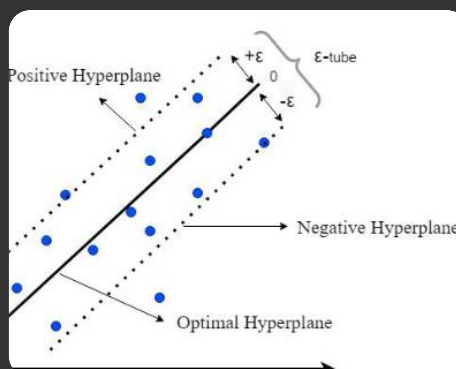
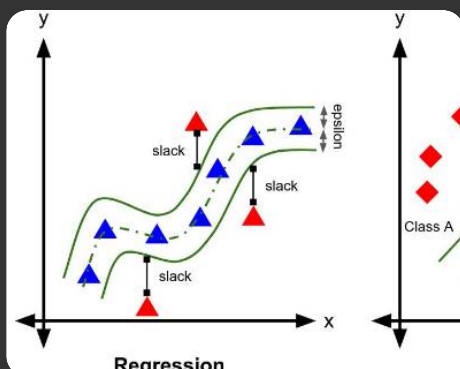
- Works for non-linear patterns

5. Random Forest Regression



- Multiple trees → average prediction
- High accuracy
- Handles complex data well

6. Support Vector Regression (SVR)



- Similar to SVM but for regression
- Uses margin of tolerance (epsilon)
- Works well for complex relationships

Quick Exam Summary

Type	Models
Classification	Logistic Regression, Decision Tree, Random Forest, SVM, KNN, Naive Bayes
Regression	Linear Regression, Polynomial Regression, Ridge/Lasso, Decision Tree, Random Forest, SVR

If you want, I can also:

- Give **real exam-style 5-mark answers**
- Provide **comparison tables (accuracy, speed, use cases)**
- Or **tell which model to use in real projects (very important for interviews)**